Transaction Fraud Detection and Risk Analysis

1. Summary

Financial fraud continues to be a major challenge for banks and payment service providers, resulting in significant financial losses and operational risks. This project presents an end-to-end fraud analytics solution that combines machine learning and business intelligence to identify suspicious transactions and uncover fraud patterns.

A synthetic transaction dataset containing 50,000 records was cleaned, transformed, and enriched through feature engineering. Additional behavioural indicators such as transaction frequency, transaction distance, high-value transactions, and customer risk patterns were created to improve analytical insights. A Random Forest classifier with SMOTE balancing was used to detect potentially fraudulent transactions, while Power BI dashboards were developed to visualize fraud trends, risk indicators, and model performance.

The analysis revealed that fraudulent transactions exhibit distinct behavioural characteristics related to transaction patterns, customer activity, and risk profiles. Interactive dashboards enabled exploration of fraud distribution across transaction types, locations, merchant categories, transaction modes, and behavioural risk levels. The solution demonstrates how machine learning and data visualization can be integrated to support fraud monitoring, risk assessment, and data-driven decision-making in financial environments.

2. Business Problem

Financial institutions process thousands of transactions every day, making manual fraud investigation inefficient and costly. Fraudulent activities often remain hidden within large volumes of transaction data and can cause significant financial and reputational damage.

The objective of this project was to build a fraud analytics system capable of identifying suspicious transactions and providing actionable insights through interactive visualizations. The solution aims to support fraud analysts by highlighting high-risk activities, monitoring transaction behaviour, and improving the efficiency of fraud detection processes.

3. Dataset Overview

Dataset Size: 50,000 transactions

3.1 Key Features:

- Transaction Amount
- Transaction Type
- Account Balance
- Device Type
- Merchant Category
- Location
- Authentication Method
- Transaction Distance
- Card Age
- Daily Transaction Count
- Fraud Label

3.2 Feature Engineering:

- Hour of transaction
- Day of transaction
- High Amount Flag
- High Distance Flag
- High Frequency Flag
- Behaviour Risk Score

4. Technical Implementation

4.1 Data Processing

- Removed missing values and duplicates
- Converted categorical and boolean fields
- Created behavioural risk indicators
- Generated time-based features

4.2 Machine Learning Pipeline

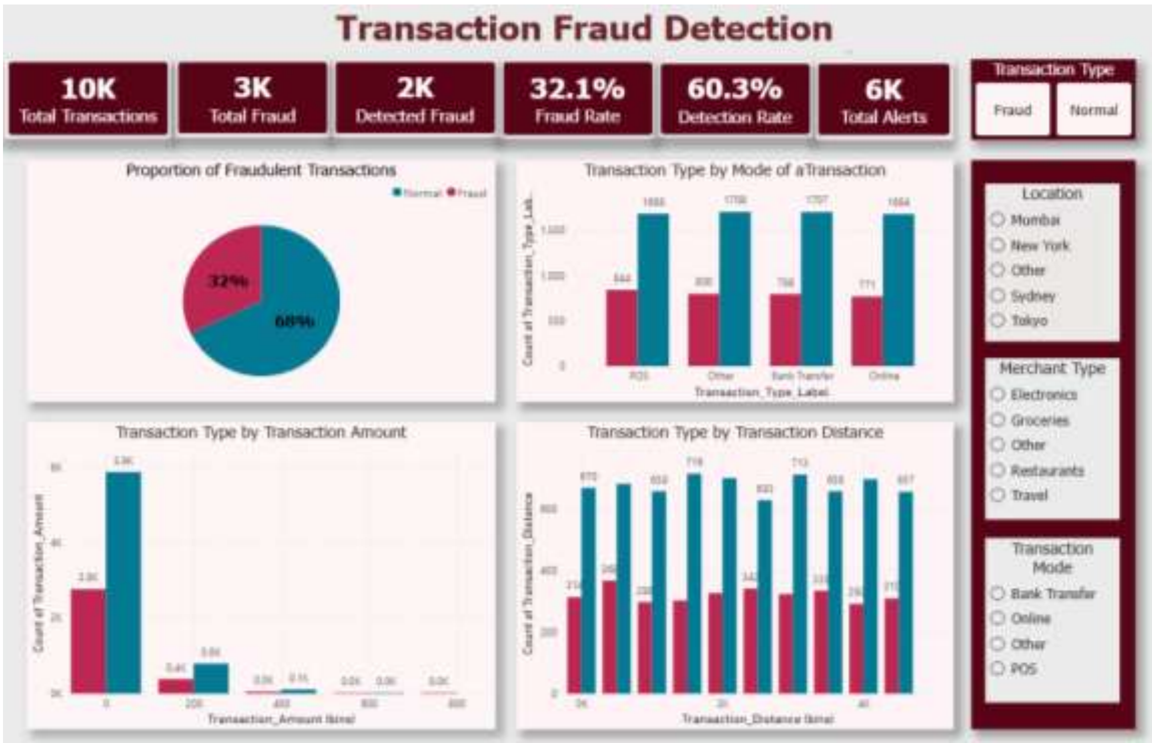
- Random Forest Classifier
- SMOTE for class balancing
- 80:20 Train-Test Split
- Threshold-based fraud prediction
- Model persistence using Joblib

5. Tools & Technologies

Category	Technology
Programming	Python
Data Analysis	Pandas, NumPy
Machine Learning	Scikit-Learn
Imbalance Handling	SMOTE
Visualization	Power BI
Database	SQL
Model Storage	Joblib

6. Dashboard Insights

6.1 Fraud Overview Dashboard



Key findings:

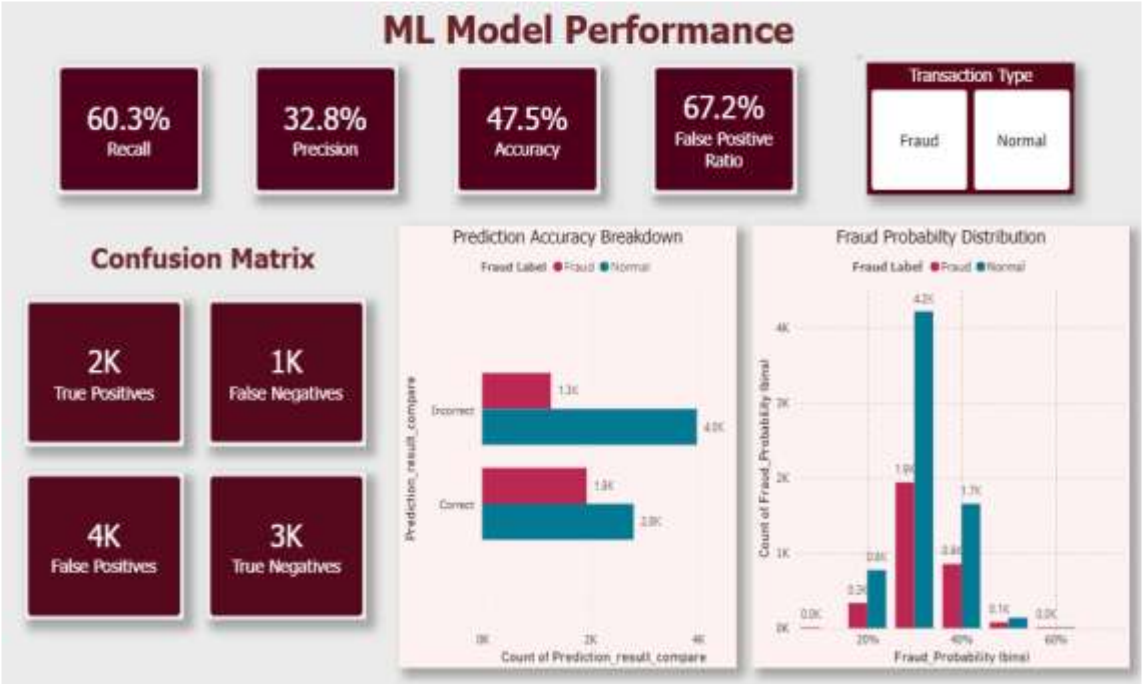
- Approximately 32% of transactions were fraudulent.
- Bank Transfer and POS transactions contributed significantly to fraud volume.
- Fraud cases were distributed across multiple transaction modes, indicating diverse attack patterns.

2. Behaviour & Risk Analysis Dashboard

Key findings:

- Higher Behaviour Risk Scores showed increased fraud rates.
- Daily transaction activity influenced fraud likelihood.
- Fraud activity remained relatively consistent throughout the day, indicating continuous fraud exposure rather than isolated peak periods.

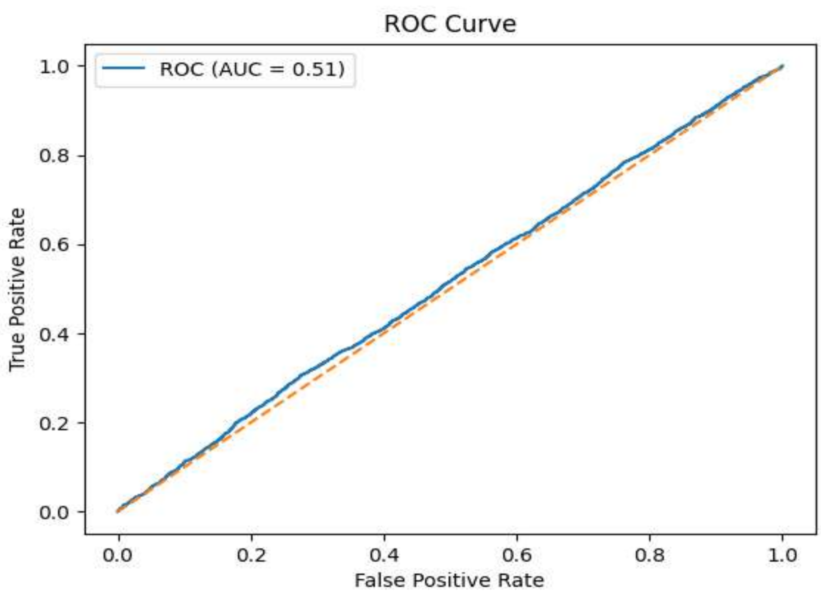
3. Model Performance Dashboard



Key findings:

- Machine learning successfully identified a large portion of fraudulent transactions.
- Fraud probability distributions helped distinguish higher-risk transactions from normal activity.
- Confusion matrix analysis provided visibility into fraud detection effectiveness.

7. ROC Curve Analysis



The ROC Curve was used to evaluate the model's ability to distinguish between fraudulent and legitimate transactions across different classification thresholds. The model achieved an AUC score of 0.51. The curve lies close to the diagonal reference line, indicating limited class separation capability. Despite this, the chosen threshold enabled the model to identify a substantial number of fraudulent transactions, supporting its use for fraud monitoring and alert generation. The ROC analysis highlights the importance of additional feature engineering and advanced modelling techniques for improving fraud detection performance.

8. Insights and recommendations

8.1 Behaviour-Based Fraud Monitoring

Observation**:** Fraud rates increase with higher Behaviour Risk Scores.

Recommendation: Prioritize transactions with high Behaviour Risk Scores by monitoring indicators such as transaction frequency, distance, and amount.

Business Impact: Improves early detection of high-risk transactions and investigation efficiency.

8.2 Channel-Wide Fraud Monitoring

Observation: Fraud occurs across Online, POS, and Bank Transfer channels.

Recommendation: Apply fraud monitoring and security controls consistently across all transaction channels.

Business Impact: Reduces fraud exposure and strengthens overall transaction security.

9. Conclusion

The Transaction Fraud Detection project successfully analysed 50,000 financial transactions and identified key fraud patterns using machine learning and interactive Power BI dashboards. The analysis revealed that approximately 32% of transactions in the dataset were fraudulent, indicating a significant level of financial risk. The machine learning model was able to detect a substantial portion of fraudulent transactions, achieving a recall of 60.3%, which makes it effective for identifying potentially suspicious activities that require further investigation.

The dashboard visualizations provided valuable insights into fraud behaviour across transaction types, transaction amounts, transaction distances, account balances, card age groups, and customer transaction patterns. The Behaviour & Risk Analysis dashboard highlighted that higher behaviour risk scores were associated with increased fraud rates, while the hourly fraud activity analysis helped identify time periods with elevated fraudulent activity. The ML Performance dashboard enabled continuous monitoring of detection results through metrics such as recall, precision, confusion matrix analysis, and fraud probability distribution.

Overall, the project demonstrates how machine learning combined with business intelligence tools can transform transaction data into actionable fraud insights, helping financial institutions strengthen risk monitoring, prioritize suspicious transactions, and support data-driven fraud prevention strategies.